

Volume threshold for adapting

- 3 byte midi control changes

Type: [1101: 4] [channel: 4]

Controller: [0] [control: 7]

Value: [0] [value: 7]

Sending messages

- 4 channel bits for opcode
- 14 bits for data

Types of data:

- Small positive floats in [0, 1]
- Positive integers
- Booleans
- Requests for operations (i.e. to reset, transfer weights)

Integers

- 14 bit unsigned integer, just clip to [0, 16384)
- Controller gets high 7 bits, Value gets low 7 bits

Floats

- Some float in $[1, 10) * 10^{(-x)}$
- Realistically, we never need lower than like $1e-8$
- 4 bits for exponent, 10 bits for significand
- Quantize the significand between [100, 999]
 - 2 bits of decimal precision

Receiving weights

- 2 bits opcode
 - START, WEIGHT, END, STATUS
- 16 bits quantized value
 - $[-1, 1](\text{float}) \Rightarrow [-32767, 32767](\text{int})$
 - Omit -32768 for symmetric range